

Anitrack

Personal Anime Watchlist & Analytics

Web Technologies I · HSBI · May 2026

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Team & responsibilities

Member	Main focus
Dengfeng Qiao	Frontend — Next.js, UI, i18n, responsive
Yurun Zhu	Backend — NestJS, MongoDB, Bee
Zhaoze Yu	API contract, tests, live demo

Problem & goals

Track anime: Planned → Watching → Completed

Monthly heatmap (GitHub-style)

Seasonal timetable & dashboard picks (local mirror)

API-first full-stack for course requirements

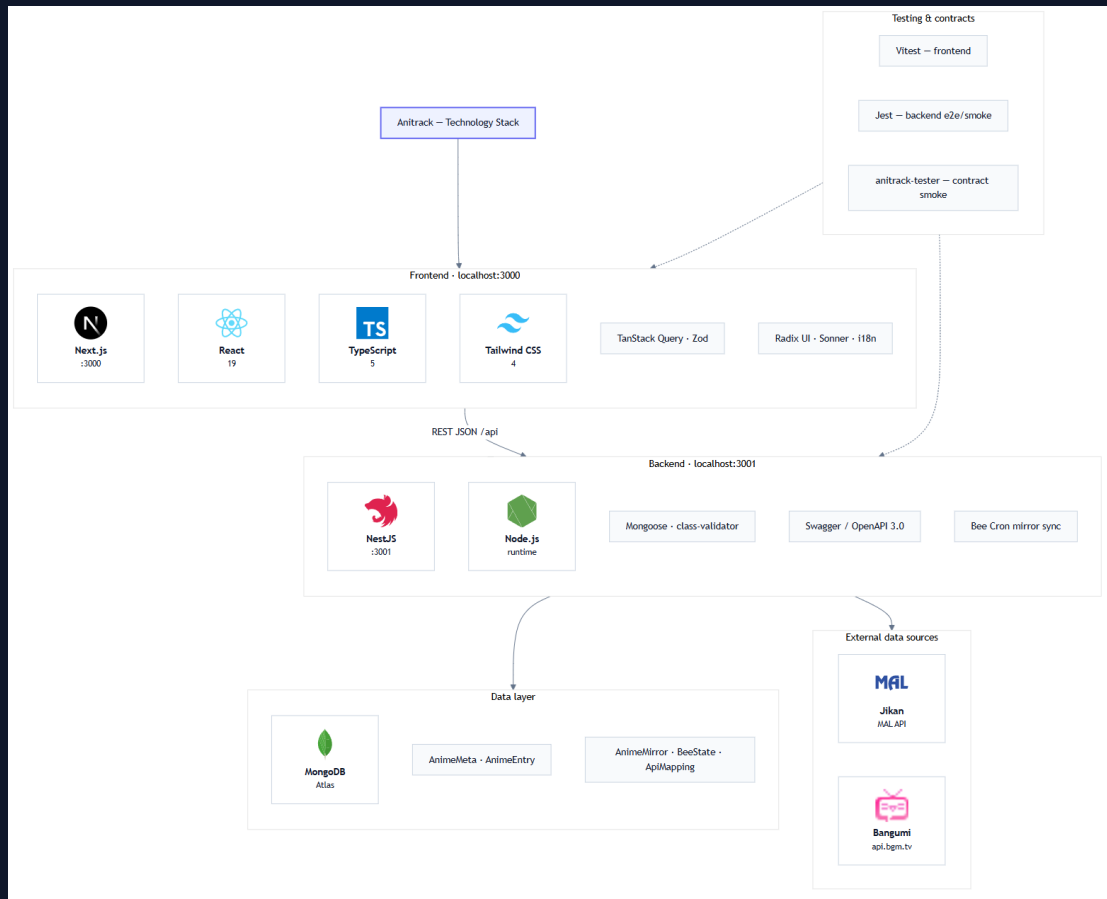
Course requirements ↔ solution

Content requirements: implemented. Organizational requirements: fulfilled in this presentation.

Requirement	Anitrack
NestJS backend	anitrack-backend :3001
CRUD HTTP API	/api/anime
API first	swagger.json, /api-docs
Tests	Jest, Vitest, contract-validator
Thin frontend	Logic in backend
Responsive	Mobile + desktop

Technology stack

*Frontend · Backend · MongoDB ·
Jikan · Bangumi*



System architecture

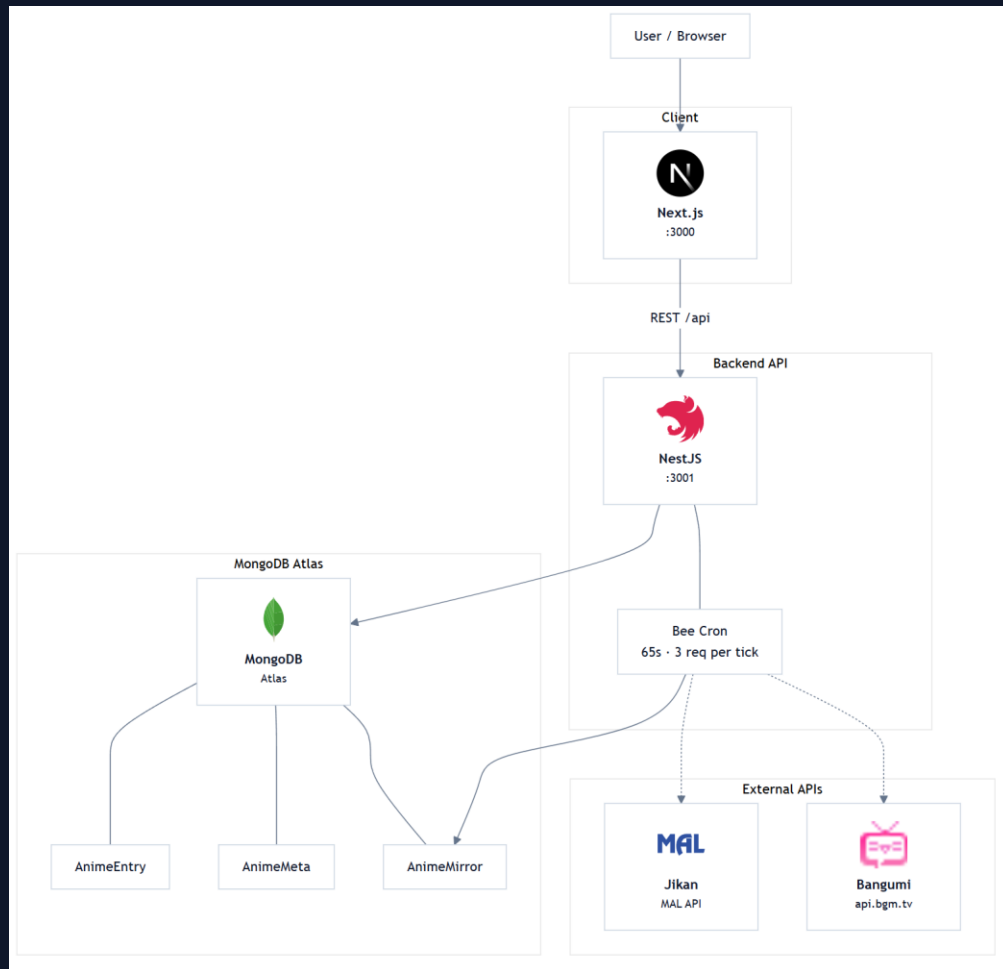
Bee mirrors Jikan/Bangumi



MongoDB; client



NestJS API only



API-first & OpenAPI

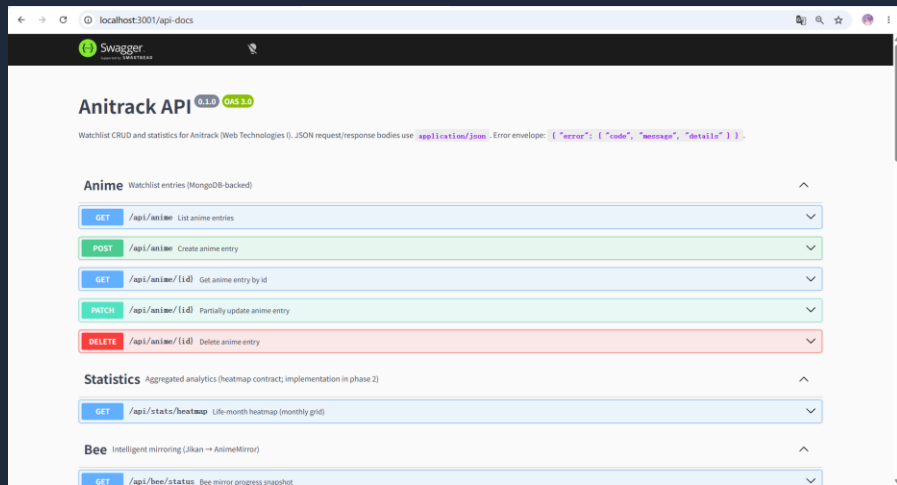
Swagger UI: <http://localhost:3001/api-docs>

Contract: anitrack-backend/swagger.json

Same API for UI + script clients (anitrack-tester)

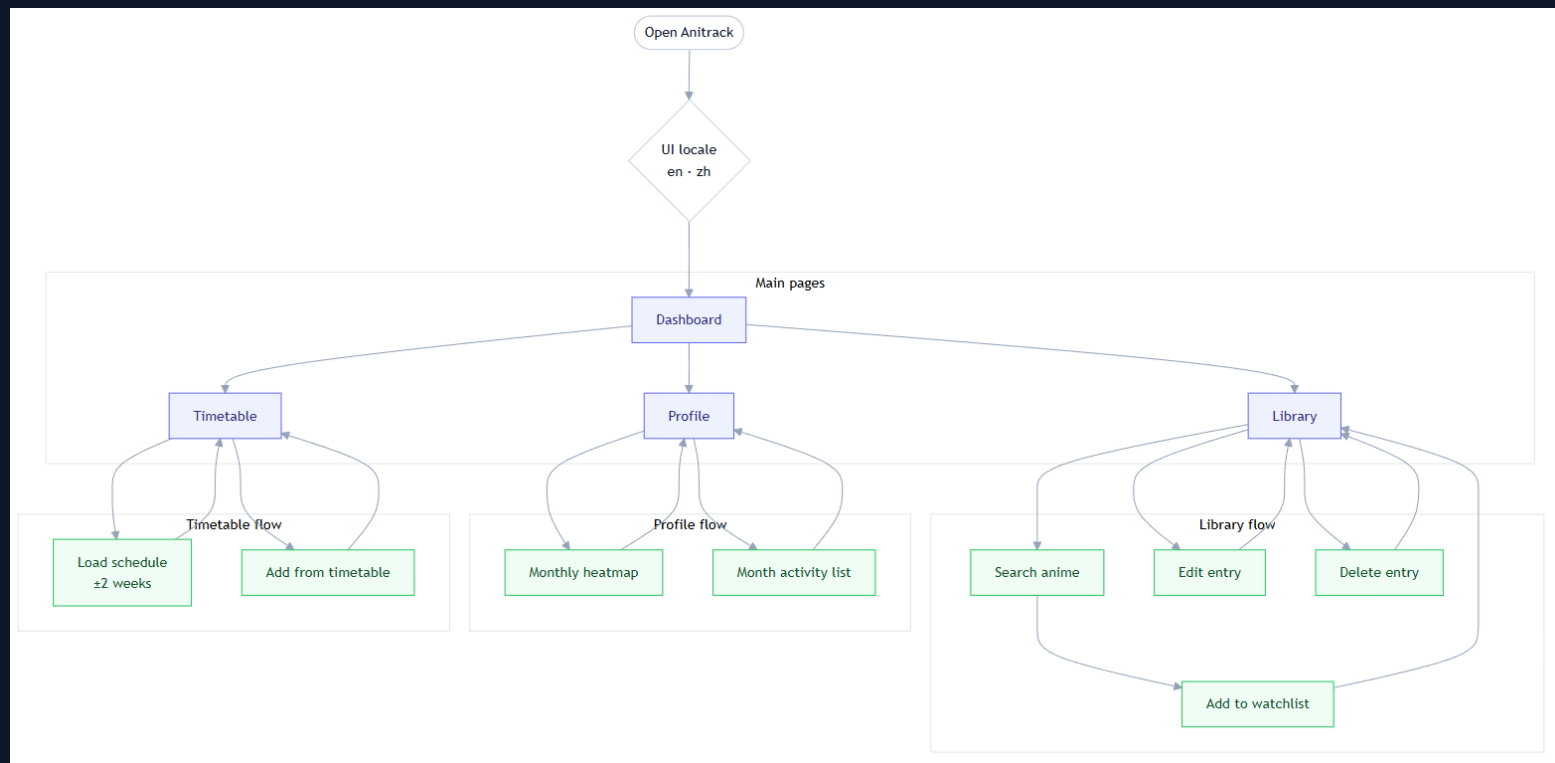
Errors: { error: { code, message, details } }

screenshot:
localhost:3001/api-docs



User flows

Dashboard · Timetable · Library · Profile

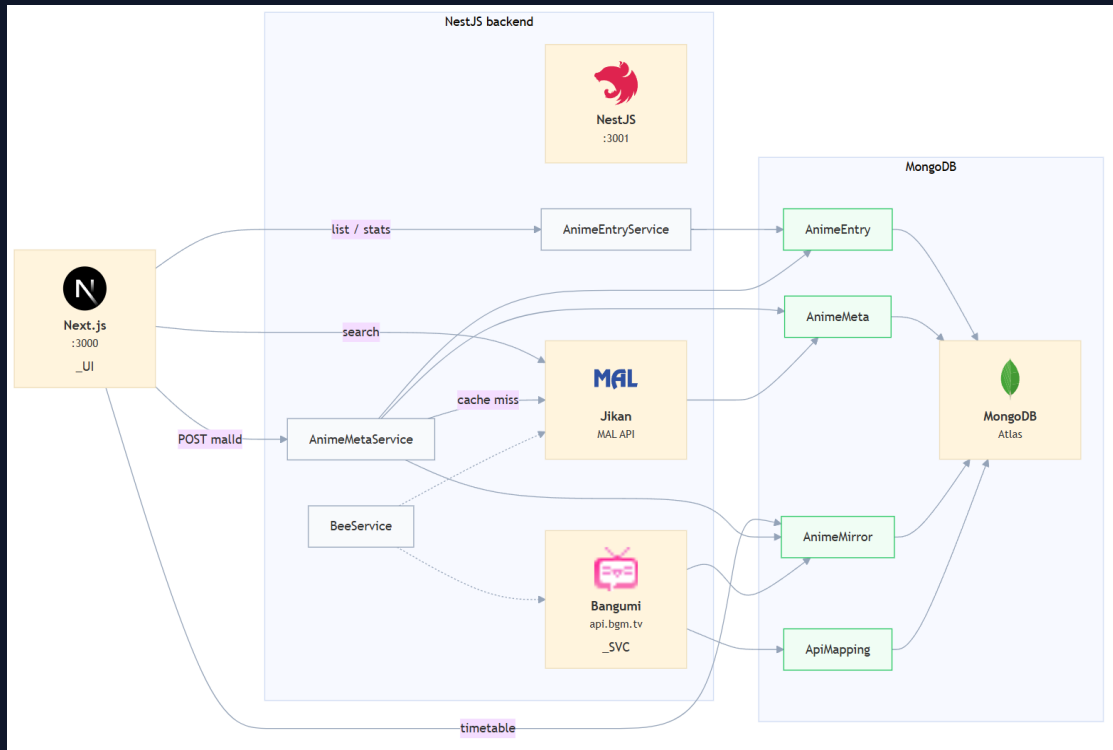


Data flow & caching

Cache-aside AnimeMeta by malId

User progress → AnimeEntry

Bee → AnimeMirror for timetable /
seasonal



Backend business logic

State machine → illegal status = HTTP 409

Heatmap aggregation on server (intensity 0–4)

Bee: 65s / 3 requests, Bangumi mapping

No business rules in React frontend

Testing strategy

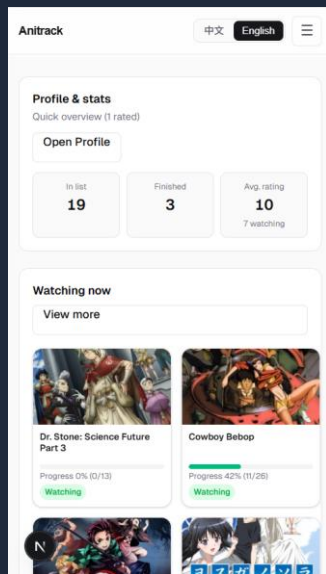
Unit: heatmap mapping (Vitest)

Integration: API + MongoDB (Jest, Vitest)

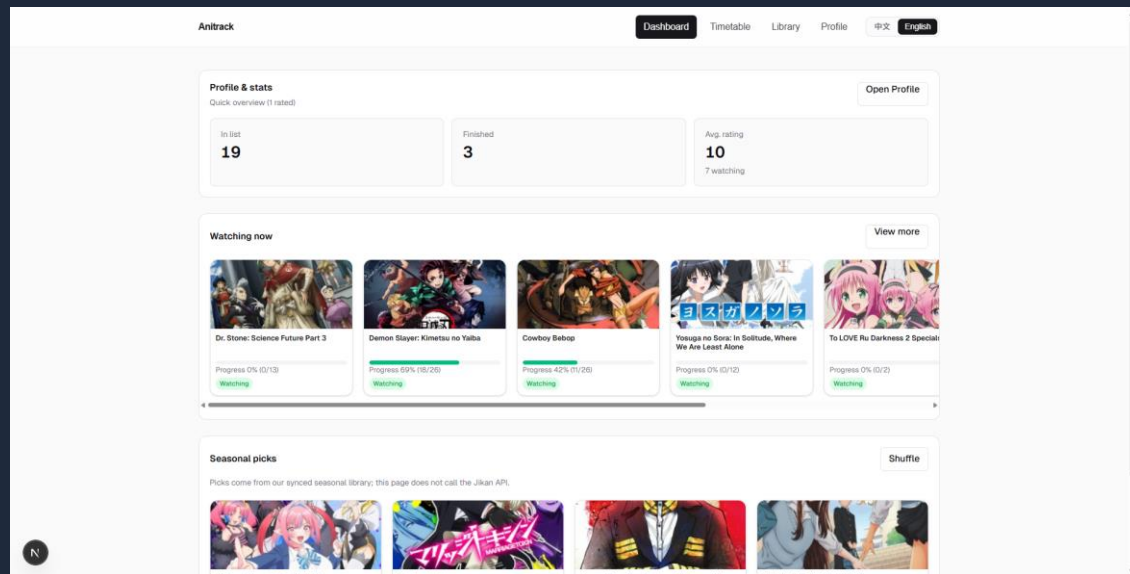
Contract: OpenAPI vs runtime (anitrack-tester)

Responsive UI

Hamburger navigation · date strip ± 2 weeks · heatmap scroll



Mobile / narrow



Desktop / wide

Live demo plan

1. Dashboard — watching & seasonal picks
2. Library — search → add to watchlist
3. Profile — heatmap → month activity
4. (Optional) Timetable — pick date

Before demo: backend :3001, frontend :3000, UI = English

Live Demo

<http://localhost:3000>

(Switch to browser — no further slides during demo)

Limitations & outlook

Timetable air times sometimes TBD (upstream data)

Single user (TEMP_USER_ID) — no auth yet

Future: multi-user, recommendations

Course core scope is complete

Thank you for your attention.

Questions?